

Experiment 32: Find the Number of Vowels in a String Using Prolog

Aim

To write and execute a **Prolog program to find the number of vowels present in a given string.**

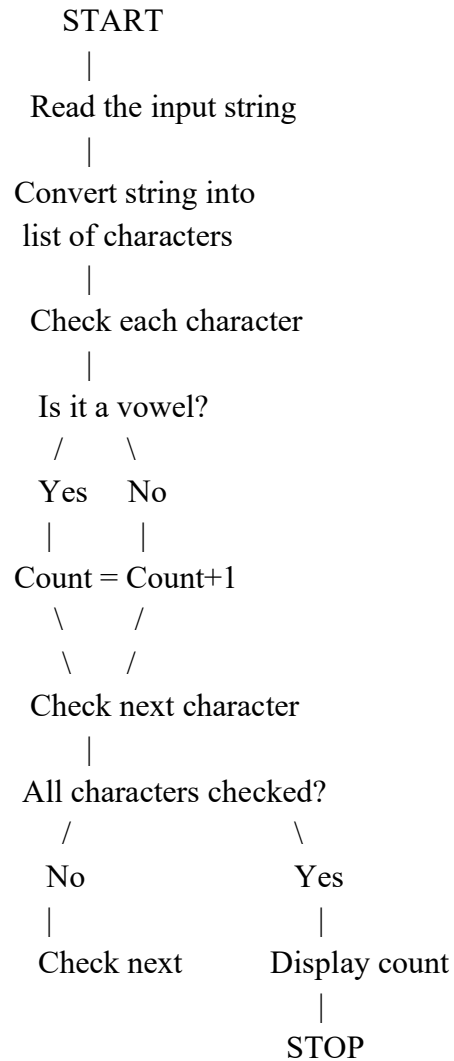
Objective

- To understand string and list processing in Prolog.
- To identify vowels from the characters of a string.
- To count and display the total number of vowels.

Algorithm

1. Start the program.
2. Define the vowels a, e, i, o, u as facts.
3. Convert the given string into a list of characters using `string_chars/2`.
4. Check each character in the list.
5. If the character is a vowel, increment the count by 1.
6. If the character is not a vowel, continue without incrementing the count.
7. Repeat until all characters are checked.
8. Display the total number of vowels.
9. Stop.

Flowchart



Prolog Code

```
vowel(a).
vowel(e).
vowel(i).
vowel(o).
vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-
    vowel(H),
```

```
count_vowels(T, C),  
Count is C + 1.
```

```
count_vowels([H|T], Count) :-  
    \+ vowel(H),  
    count_vowels(T, Count).
```

```
vowels(String, Count) :-  
    string_chars(String, Chars),  
    count_vowels(Chars, Count).
```

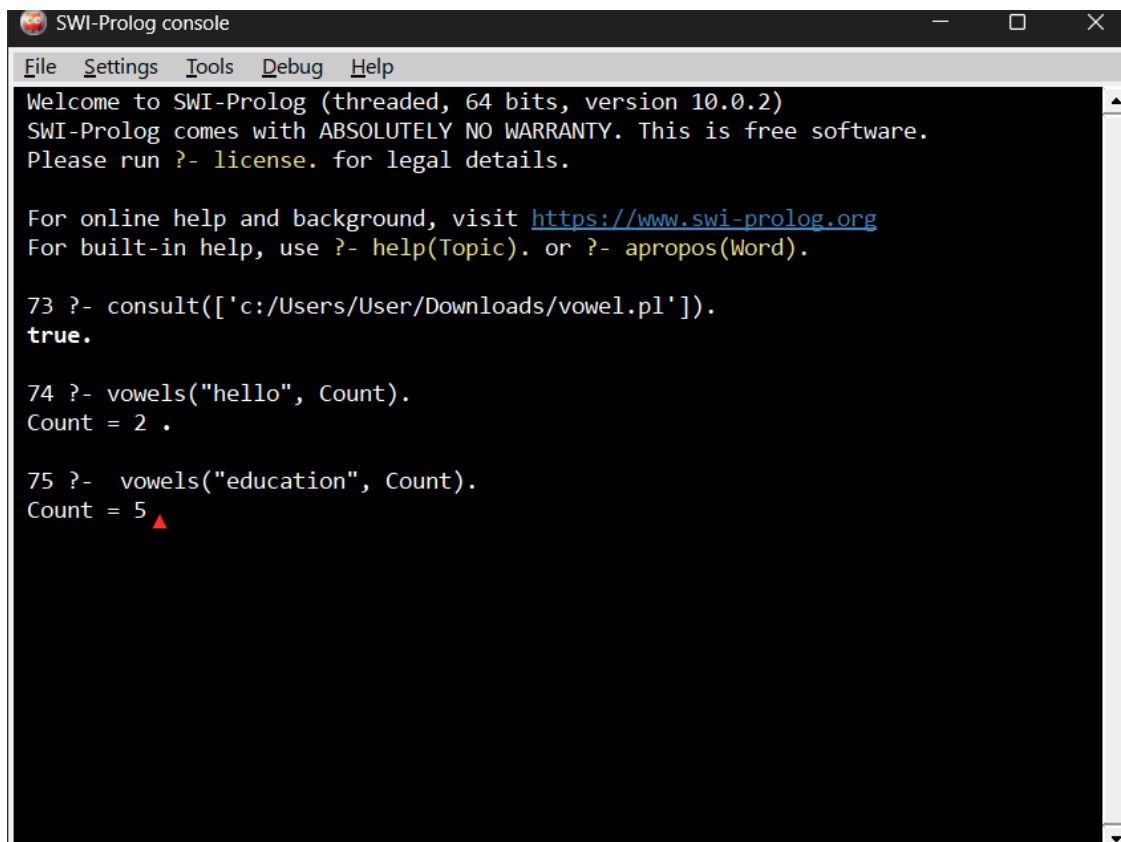
Sample Output

Query 1:

```
?- vowels("hello", Count).
```

Query 2:

```
?- vowels("education", Count).
```



```
SWI-Prolog console  
File Settings Tools Debug Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
73 ?- consult(['c:/Users/User/Downloads/vowel.pl']).  
true.  
  
74 ?- vowels("hello", Count).  
Count = 2 .  
  
75 ?- vowels("education", Count).  
Count = 5 ▲
```

Result

Thus, the Prolog program to **find the number of vowels in a given string** was successfully implemented and executed.

Conclusion

The experiment demonstrates how Prolog can process strings using **recursion, facts, rules, and list processing**. The program successfully identifies vowels and calculates their total count in the given string.